Large Data Set: OCR MEI (Health)

Questions 1–14

Question 1. These exercises were written using LDS6 but should continue to be of use for future Health Large Data Set samples. State the following facts about the Large Data Set:

a) The age of the youngest person.
b) The age of the oldest person.
c) The six categories of marital status.
d) What the category "Food30" is recording.
e) What the category "Arm" is recording.
f) The country in which the original survey is conducted.

Question 2. State the units of the following:

a) Weight.
b) Height.
c) Body mass index.
d) Lengths of body parts.
e) Pulse rate.
f) Blood pressure.

Question 3. What is the data set taken from?

Question 4. Explain the difference between the systolic blood pressure and diastolic blood pressure.

Question 5. Explain why there are columns in the Large Data Set labelled with a 1, 2, 3, 4, in the context of what was happening on the day that the data was collected.

Question 6. What data do the columns labelled Systolic and Diastolic represent in the Large Data Set?

Question 7. Mustafa wants a sample of 20 people from the Large Data Set.
He claims that all he needs to do is pick 20 random numbers from between 1 and 200 and use those rows from the spreadsheet.

a) State what type of sampling he has chosen
b) Explain why this might not end up with 20 useful sets of data.

Question 8. Kristina would like to clean the data set before she samples from it. She decides to remove any rows which contain a #N/A except the 'Systolic4' and the 'Distolic4' columns. Explain the effect of this on the data set.

Question 9. Katie would like to take a sample to do a comparison of body mass index for different categories of marital status. She would like a sample of 50 with each marital status proportionally represented.

a) State the type of sampling she should use.
b) Explain how she would do this.

Question 10. The table below shows a sample of 10 people from the Large Data Set.

Sex	Height
Female	173.4
Female	171.4
Male	183.5
Female	174.7
Male	180.1
Male	184.9
Male	186.2
Male	173.3
Male	159.4
Male	188.4

a) Find the mean and the standard deviation of the height of this sample.
b) Give a reason why the sample used above may have biased data.
c) Any value more than 2 standard deviations from the mean can be identified as an outlier. Find any outliers in this sample of heights.

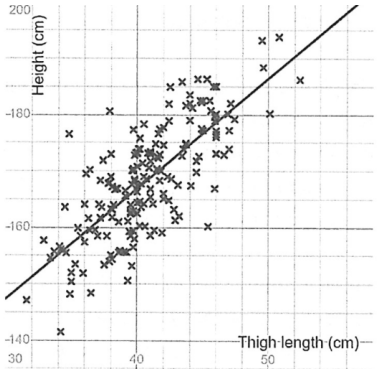
The mean height for the whole Large Data Set is 168.0 cm^3 and the standard deviation is 9.9 cm^3.

d) Give one comparison of central tendency for this sample vs the full data set.
e) Give one comparison of spread of this sample vs the full data set.

Question 11. The scatter graph below shows the height against thigh length of the 179 data points in the Large Data Set. A regression line has been found, given by

$$h = 1.95t + 88.70$$

where h is the height of a person in cm and t is the thigh length of a person in cm.



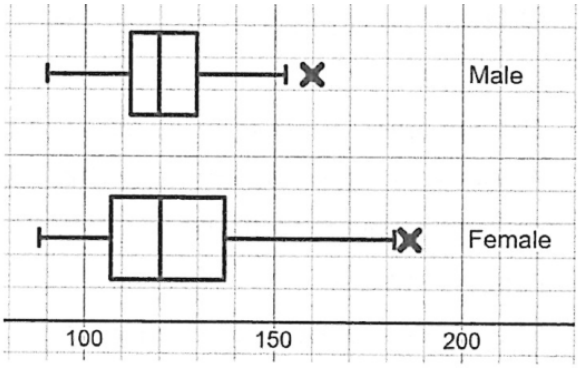
One of the people not included in the data set is 116.3 cm tall.

a) Assuming you can make a prediction by using the regression line, calculate the thigh length of this person, giving your answer to 1 decimal place.

The product moment correlation coefficient for the data is 0.7518

b) Explain why it is reasonable to model the relationship between t and h as a linear graph.

Question 12. The average diastolic blood pressure for males and females is shown in the boxplots below.

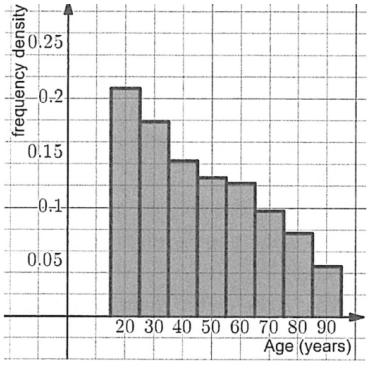


a) Estimate the median of males and the median of females.

b) Estimate the interquartile range of males and the interquartile range of females.

c) Use your answers to compare the average diastolic blood pressure for males and females in the Large Data Set.

Question 13. The histogram below shows the ages, in years, of the people in the Large Data Set.



The heights of the bars are $\{0.209, 0.179, 0.143, 0.128, 0.122, 0.097, 0.076, 0.046\}$

a) Which class interval contains the median?

b) Estimate the mean.

An adult is selected at random.

c) Find the probability that they are younger than 45 years old.

d) Explain why the final bar may not be a suitable width to represent the data.

Question 14. Prerequisite knowledge for Q14: PMCC Hypothesis Testing Emily believes that there is a positive correlation between BMI and average systolic blood pressure. She collects a random sample of 20 people from the Large Data Set and calculates the sample's correlation coefficient to be 0.8159. The critical value for $n = 20$ for a one-tailed test at a 5% significant level is 0.3783

a) Carry out a hypothesis test at the 5% significance level to investigate whether Emily's belief is supported.

Emily's teacher is suspicious of these results, and they create a scatter graph for the whole data set, drawing a vertical and horizontal line going through the mean point. They state that "there are approximately the same number of data points in each quarter of the graph."

b) Explain why the teacher's graph shows and why they believe Emily selected the data with bias.